

Secondary Examination, 2019

Mathematics

Time-3 $\frac{1}{4}$ Hours

M.M- 80

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) Candidate must write first his/ her Roll No. on the question paper compulsorily.
- 2) All the questions are compulsory.
- 3) Write the answer to each question in the given answer- book only.
- 4) For questions having more than one part the answers to those parts are to be written together in continuity.
- 5) If there is any error/difference/contradiction in Hindi and English versions of the question paper of Hindi version be treated valid.
- 6)

Section	Q.Nos.	Marks per question
A	1-10	1
B	11-15	2
C	16-25	3
D	26-30	6
- 7) There are internal choices in Q. Nos. 28 and 30.
- 8) Write on both sides of the pages of your answer-book. If any rough work is to be done, do it on last pages of the answer-book and cross with slant lines and write ' Rough Work' on them.
- 9) Draw the graph of Q. No. 26 on the graph paper.

PART-A

Question.1 Find the square of 62 by Dwandwa yoga method.

Solution.

Simplify the expression,

Dwandwa of 6 / dwandwa of 62 / dwandwa of 2

$$= 36 / 2 \times 6 \times 2 / 2^2$$

$$= 36 / 24 / 4$$

$$= 3844$$

Hence, the value of 3844.

Question.2 Solve $(x+1)(x+2) = (x-5)(x-6)$.

Solution.

Simplify the expression,

$$(x+1)(x+2) = (x-5)(x-6)$$

$$a = 1, b = 2, c = -5, d = -6$$

$$x = \frac{cd - ab}{a + b - c - d}$$

$$x = \frac{30 - 2}{3 - (-11)} = \frac{28}{14} = 2$$

Hence, the value is 2.

Question.3 Find the HCF of 68 and 119.

Solution.

Simplify the expression,

$$68 = 2^2 \times 17$$

$$119 = 7 \times 17$$

The common value is HCF $(68, 119) = 17$

Hence, the value is 17.

Question.4 Find the value of $\tan^2 60^\circ + 3\cos^2 30^\circ$

Solution.

Simplify the expression,

$$\begin{aligned} & \tan^2 60^\circ + 3\cos^2 30^\circ \\ &= (\sqrt{3})^2 + 3\left(\frac{\sqrt{3}}{2}\right)^2 \\ &= 3 + \frac{9}{4} = \frac{12+9}{4} = \frac{21}{4} \\ &= \frac{21}{4} \end{aligned}$$

Hence, the value of $\frac{21}{4}$

Question.5 If $\sin 2A = \cos(A - 18^\circ)$, find the value of A.

Solution.

Simplify the expression,

$$\begin{aligned} & \cos(90 - 2A) = \cos(A - 18) \\ & 90 - 2A = A - 18 \\ & -2A - A = -18 - 90 \\ & -3A = -108 \\ & A = 36^\circ \end{aligned}$$

Hence, the value of $A = 36^\circ$.

Question.6 Write the locus of the centre of rolling circle in a plane.

Solution.

Question.7 If $\triangle ABC \sim \triangle DEF$ in which $AB=1.6\text{cm}$, and $DE = 2.4\text{cm}$. Find the ratio of the areas of $\triangle ABC$ and $\triangle DEF$.

Solution.

Simplify the expression,

$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle DEF)} = \left(\frac{AB}{DE}\right)^2$$

$$= \left(\frac{1.6}{2.4}\right)^2 = \left(\frac{4}{9}\right)$$

$$\Rightarrow 4:9$$

Hence, the value of ratio 4:9

Question.8 Two players a chess match. It is given that probability of winning the match by A is $\frac{5}{6}$. Find the probability of winning the match by B.

Solution.

Simplify the expression,

$$p(A) + p(B) = 1$$

$$\frac{5}{6} + p(B) = 1$$

$$p(B) = 1 - \frac{5}{6}$$

$$p(B) = \frac{6-5}{6}$$

$$p(B) = \frac{1}{6}$$

Hence, the value of $p(B) = \frac{1}{6}$.

Question.9 If fare of a car for first kilometer is Rs-20 and for after 1 kilometer is Rs-11, then the total fare for 15 kilometers.

Solution.

Simplify the expression,

$$a_1 = 20$$

$$d = 11$$

$$n = 15$$

$$a_n = a + (n-1) \cdot d$$

$$= 20 + (15-1) \cdot 11$$

$$= 20 + 14 \times 11$$

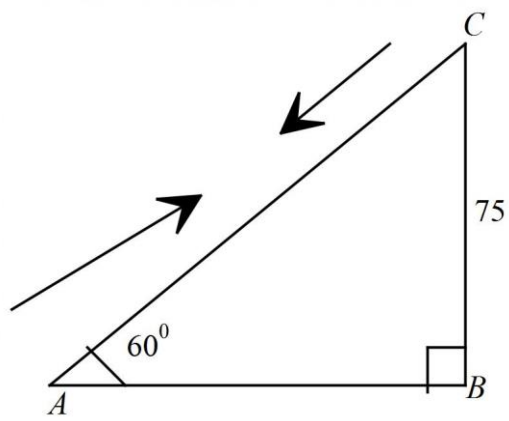
$$= 20 + 154$$

$$a_n = 174$$

Hence, the value of $a_n = 174$.

Question.10 A kite is flying at a height of 75 metres from the level of ground attached to a string inclined at 60° to the horizontal. Find the height of the string.

Solution.



Let $AC = x$

Simplify the expression,

$$\sin 60^\circ = \frac{75}{x}$$

$$\frac{\sqrt{3}}{2} = \frac{75}{x}$$

$$\sqrt{3}x = 75 \times 2$$

$$\sqrt{3}x = 150$$

$$x = \frac{150}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{150\sqrt{3}}{3} = 50\sqrt{3}$$

Hence, the height of kite is $50\sqrt{3}$.

PART -B

Question.11 Find the cube of 42 by using Upsutra Anurupyena.

Solution.

Simplify the expression,

(i)	(ii)	(iii)	(iv)
4^3	$4^2 \times 2 = 32$	4×2^2	2^3
+64	+64	16	8
		+32	
+64	96	48	8

$$= 77088$$

Hence, the value of 77088.

Question.12 Prove that $7\sqrt{5}$ is an irrational number.

Solution.

Simplify the expression,

Let $7\sqrt{5}$ is a rational number.

Any rational number can be written in to the from of a/b , $b \neq 0$

$$7\sqrt{5} = a/b$$

$$\sqrt{5} = a/7b \quad (\text{rational number})$$

Then, $\sqrt{5}$ also a rational number.

Hence, we can say $7\sqrt{5}$ is a irrational number.

Question.13 Radius of a circle is 9cm and the angle of the sector is 70° . Find the area of the minor sector of the circle. $\left(\pi = \frac{22}{7}\right)$.

Solution.

Simplify the expression,

$$r = 9\text{cm}$$

$$\theta = 90^\circ$$

We know that,

Area of minor sector

$$= \frac{\pi r^2 \theta}{360^\circ}$$

$$= \frac{22}{7} \times 9 \times 9 \times \frac{90}{360}$$

$$= \frac{81 \times 11}{14}$$

$$= \frac{891}{14}$$

Hence, the value of $\frac{891}{14}$.

Question.14 The height of a cylinder is 21cm and its curved surface area is 924cm^2 . Find the radius of cylinder.

Solution.

Simplify the expression,

We know that, the surface are of cylinder.

$$= 2\pi rh$$

$$\text{Given } h = 21\text{cm}$$

$$= 2\pi rh$$

$$924 = 2 \times \frac{22}{7} \times r \times 21$$

$$r = \frac{924 \times 7}{2 \times 22 \times 21}$$

$$r = 7\text{cm}$$

Hence, the radius of cylinder is $r = 7\text{cm}$.

Question.15

Solution.

Simplify the expression,

$$A \rightarrow B = 125\text{km}$$

$$\text{Speed of car} = 50\text{km} / h$$

$$T = \frac{125}{50} = 2.5\text{hour}$$

AP 1, 2, 3,8

$$a_1 = 1$$

$$d = 2 - 1 = 1$$

$$n = 8$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{8}{2} [2 \times 1 + (8-1) \cdot 1]$$

$$= 4[2 + 7]$$

$$= 36 \text{ minutes}$$

$$= 2.5 \text{ hour} + 36 \text{ minutes}$$

$$= 3 \text{ hour } 6 \text{ minutes}$$

$$= 3:06 \text{ minutes}$$

Hence, the value of 3:06 minutes.

Question.16 Solve the equation $\frac{1}{x-2} + \frac{1}{x-1} = \frac{6}{x}$ where $x \neq 1, 2$ by factorization method.

Solution.

Simplify the expression,

$$\frac{x-1+x-2}{(x-2)(x-1)} = \frac{6}{x}$$

$$\frac{2x-3}{(x-2)(x-1)} = \frac{6}{x}$$

$$2x^2 - 3x = 6(x^2 - x - 2x + 2)$$

$$2x^2 - 3x = 6x^2 - 6x - 12x + 12$$

$$= 2x^2 - 6x^2 - 3x + 18x - 12 = 0$$

$$= -4x^2 + 15x - 12 = 0$$

$$= 4x^2 - 15x + 12 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-15) \pm \sqrt{(-15)^2 - 4 \times 4 \times 12}}{2 \times 4}$$

$$x = \frac{15 \pm \sqrt{-33}}{8}$$

Hence, the value of $x = \frac{15 \pm \sqrt{-33}}{8}$.

Question.17 Find the sum of all natural numbers divisible by 5 between 2 and 101.

Solution.

Simplify the expression,

$$d = a_3 - a_2 = 5 - 3 = 2$$

$$a_2 = a + d \Rightarrow 3$$

$$a_3 = a + 2d = 5$$

$$d = 2$$

$$a + 2 = 3$$

$$a = 3 - 2 = 1$$

$$S_{20} = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{20}{2} [2 \times 1 + (20-1)2]$$

$$= 10 [2 + 38] = 10 \times 40 = 400$$

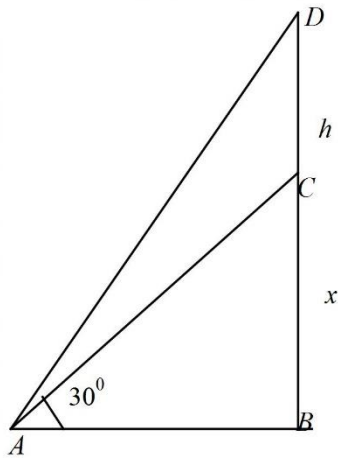
$$400 = 20^2$$

Hence, the value of 400.

Question.18 From a point on the ground which is 120m away from the foot of the unfinished tower, the angle of elevation of the top of the tower is found to be 30° . Find how much height of tower to increase so that its angle of elevation at same point become 30° ?

Solution.

Simplify the expression,



$\triangle ABC$

$$\tan 30^\circ = \frac{p}{b}$$

$$\frac{1}{\sqrt{3}} = \frac{x}{120}$$

$$\sqrt{3}x = 120$$

$$x = \frac{120}{\sqrt{3}}$$

$$x = \frac{120}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 40\sqrt{3}$$

$$x = 40\sqrt{3}$$

$\triangle ADC$

$$\tan 60^\circ = \frac{p}{b}$$

$$\frac{\sqrt{3}}{1} = \frac{x+y}{120}$$

$$x+y = 120\sqrt{3}$$

$$40\sqrt{3} + y = 120\sqrt{3}$$

$$y = 120\sqrt{3} - 40\sqrt{3}$$

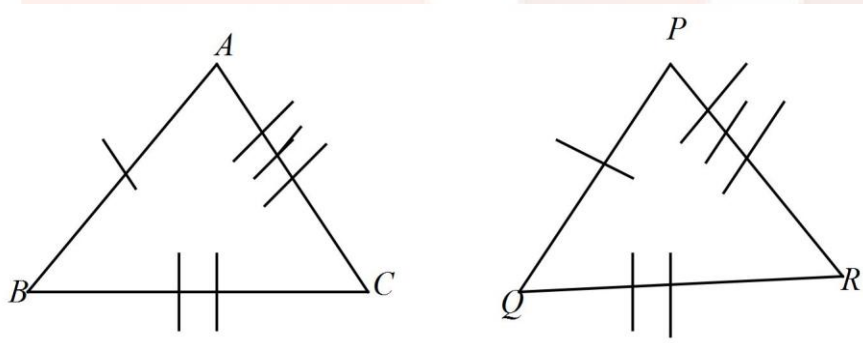
$$y = 80\sqrt{3}m$$

Hence, the increased the height of tower $y = 80\sqrt{3}m$.

Question.19 Prove that if area of two similar triangles are equal, then they are congruent.

Solution.

Prove the expression,



$$\triangle ABC \sim \triangle PQR$$

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR}$$

$$\left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2 = \frac{1}{1}$$

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{1}{1}$$

$$\frac{AB}{PQ} = \frac{1}{1}$$

relation (i) and (iv)

$$AB = PQ$$

(ii) and (iv)

$$BC = QR$$

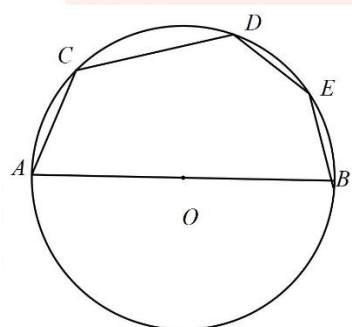
SSS

$$\triangle ABC \cong \triangle PQR$$

Hence, prove it.

Question.20 In fig. AOB is the diameter of a circle and C,D and E are three points situated on semi-circle. Find the value of $\angle ACD + \angle BED$.

Solution.



$$\angle ACD + \angle BED = ?$$

$$\angle ACB = 90^\circ \dots\dots(i)$$

$ABCD$ is a cyclic quadrilateral

$$\angle ACD + \angle AED = 180^\circ \dots\dots(ii)$$

$$\angle ACD + \angle BED$$

$$= \angle ACD + \angle AED + \angle AEB$$

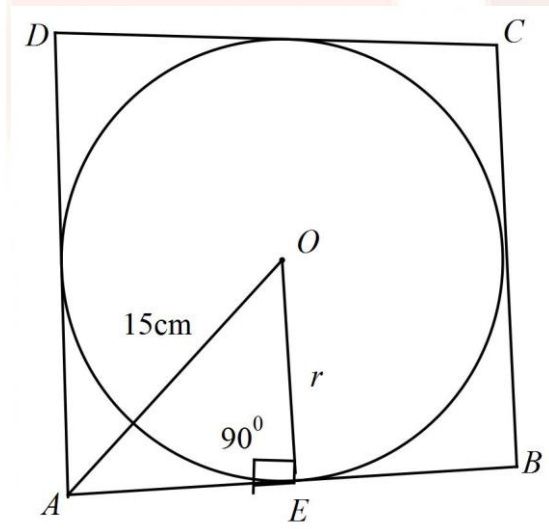
$$\Rightarrow \angle ACD + \angle BED = 180^\circ + 90^\circ = 270^\circ$$

$$\Rightarrow \angle ACD + \angle BED = 270^\circ$$

Hence, the value is of angle $\angle ACD + \angle BED = 270^\circ$.

Question.21 A circle with centre 'O' touches all the four sides of a quadrilateral $ABCD$ internally in such a way that it divides AB in 3:1 and $AB = 12\text{cm}$ then find the radius of circle where $OA = 15\text{cm}$.

Solution.



$$AE : EB = 3 : 1$$

$$AB = 12\text{cm}, \quad OA = 15\text{cm}$$

$$\frac{AE}{AB} = \frac{3}{4}$$

$$\frac{AE}{12} \times 12 = \frac{3}{4} \times 12$$

$$AE = 9\text{cm}$$

$\triangle AOE$

$$AE^2 + OE^2 = OA^2$$

$$9^2 + r^2 = 15^2$$

$$r^2 = 225 - 81$$

$$r^2 = 144$$

$$r = \sqrt{144}$$

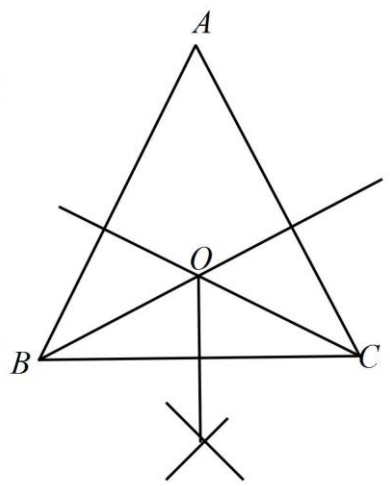
$$r = 12\text{cm.}$$

Hence, the value is $r = 12\text{cm.}$

Question.22 Construct an incircle of an equilateral triangle with side 5cm.

Solution.

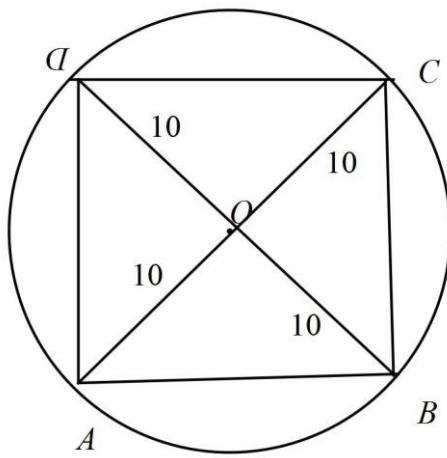
Consider



$$r = OD$$

Question.23 Find the area of the square inscribed in a circle of radius 10cm.

Solution.



Simplify the expression,

$$r = 10\text{cm}$$

$$= \frac{1}{2} \times 10 \times 10$$

$$= AB^2 + AD^2 = BD^2$$

$$x^2 + x^2 = 20^2$$

$$2x^2 = 400$$

$$x^2 = 200\text{cm}^2$$

Hence, the area of square inscribed $x^2 = 200\text{cm}^2$.

Question.24 A sphere of 6cm diameter is dropped into cylindrical vessel of diameter 12cm. Find the rise in water in the vessel.

Solution.

Simplify the expression,

Radius of sphere=3cm

Radius of cylinder vessel =6cm

$$\pi r_1^2 h = \frac{4}{3} \pi r_2^3$$

$$6 \times 6 \times h = \frac{4}{3} \times 3 \times 3 \times 3$$

$$h = \frac{4 \times 3 \times 3}{6 \times 6}$$

$$h = 1\text{cm}$$

Hence, the value of $h = 1\text{cm}$.

Question.25 A bag contains 15 cards. The numbers 1,2,3,4....15 are printed on them. A card random drawn from the bag. Find the probability that the number on the card is.

- (i) A prime number
- (ii) A number is divisible by 2

Solution.

Simplify the expression,

$$s = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$\text{Indivisible number} = \{2, 3, 5, 7, 11, 13\}$$

$$(i) \text{probability of indivisible number} = \frac{6}{15} = \frac{2}{5}$$

$$(ii) \text{A number of divisible by 2} = \{2, 4, 6, 8, 10, 12, 14\}$$

$$= \frac{7}{15}$$

Hence, the value of case (i) in $\frac{2}{5}$ and case (ii) in $\frac{7}{15}$.

26. Solve the following pair of linear equation by graphical method.

$$3x - 5y = -1$$

$$2x - y = -3$$

Thus find the value of A in the relation $(x + y)^2 = A$

Solution:

Consider

$$3x - 5y = -1 \text{-----(i)}$$

$$2x - y = -3 \text{-----(ii)}$$

x	3	8
y	2	5

x	0	1
y	3	5

Putting $y = 2$ in equation (i)

$$3x - 5y = -1$$

$$3x - 10 = -1$$

$$3x = -1 + 10$$

$$x = \frac{9}{3}$$

$$x = 3$$

Putting $y = 5$ in equation (i)

$$3x - 5y = -1$$

$$3x - 5 \times 5 = -1$$

$$3x = -1 + 25$$

$$3x = 24$$

$$x = 8$$

Again, putting $x = 0$ in equation (ii)

$$2x - y = -3$$

$$2(0) - y = -3$$

$$-y = -3$$

$$y = 3$$

Putting $x = 1$ in equation (ii)

$$2x - y = -3$$

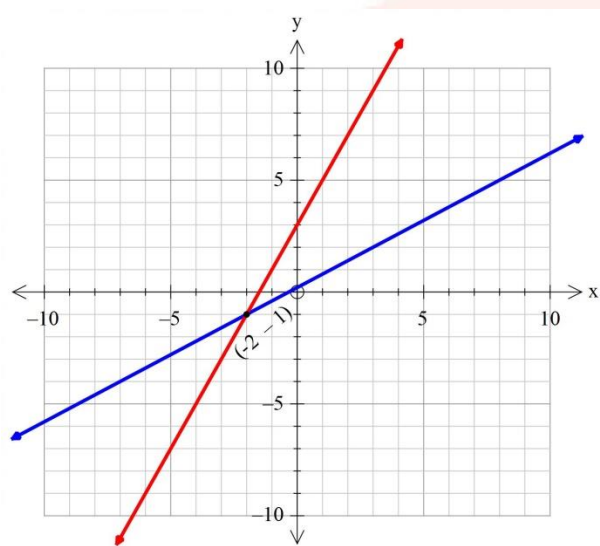
$$2 \times 1 - y = -3$$

$$2 - y = -3$$

$$-y = -3 - 2$$

$$-y = -5$$

$$y = 5$$



Then,

$$(x + y)^2 = A$$

$$(-2 - 1)^2 = A$$

$$A = 9$$

27. Prove that:

$$(i) \frac{1 + \sec A}{\sec A} = \frac{\sin^2 A}{1 - \cos A}$$

Solution:

Consider

$$\begin{aligned} LHS &= \frac{1 + \sec A}{\sec A} \\ &= \frac{\left(1 + \frac{1}{\cos A}\right) \cos A}{\left(\frac{1}{\cos A}\right) \cos A} \\ &= \frac{\cos A + 1}{1} \times \frac{1 - \cos A}{1 - \cos A} \\ &= \frac{1 - \cos^2 A}{1 - \cos A} \\ &= \frac{\sin^2 A}{1 - \cos A} \quad \left(\begin{array}{l} \because \sin^2 A + \cos^2 A = 1 \\ \therefore \sin^2 A = \cos^2 A - 1 \end{array} \right) \\ &= RHS \end{aligned}$$

$$(ii) \frac{\cos^2 \theta}{1 - \tan \theta} = \frac{\sin^3 \theta}{\sin \theta - \cos \theta} = 1 + \sin \theta \cos \theta$$

Solution:

Consider

$$\begin{aligned}
 LHS &= \frac{\cos^2 \theta}{1 - \tan \theta} + \frac{\sin^3 \theta}{\sin \theta - \cos \theta} \\
 &= \frac{\cos^2 \theta}{\left(1 - \frac{\sin \theta}{\cos \theta}\right)} + \frac{\sin^3 \theta}{\sin \theta - \cos \theta} \\
 &= \frac{\cos^2 \theta}{\left(\frac{\cos \theta - \sin \theta}{\cos \theta}\right)} + \frac{\sin^3 \theta}{\sin \theta - \cos \theta} \\
 &= \frac{\cos^3 \theta}{\cos \theta - \sin \theta} - \frac{\sin^3 \theta}{\cos \theta - \sin \theta} \\
 &= \frac{\cos \theta - \sin \theta (\cos^2 \theta + \sin \theta \cos \theta + \sin^2 \theta)}{\cos \theta - \sin \theta} \quad (\because a^3 - b^3 = (a - b)(a^2 + ab + b^2)) \\
 &= 1 + \sin \theta \cos \theta \\
 &= RHS
 \end{aligned}$$

Hence, LHS = RHS Proved.

OR,

Prove that:

$$(i) \frac{\sin \theta}{(1 + \cos \theta)} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

Solution:

Consider

$$\begin{aligned}
 LHS &= \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \\
 &= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta) \sin \theta} \\
 &= \frac{\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta}{(1 + \cos \theta) \sin \theta} \\
 &= \frac{2 + 2 \cos \theta}{(1 + \cos \theta) \sin \theta} \\
 &= \frac{2(1 + \cos \theta)}{(1 + \cos \theta) \sin \theta} \\
 &= 2 \operatorname{cosec} \theta \\
 &= RHS
 \end{aligned}$$

Hence,

$$(ii) \frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$$

Solution:

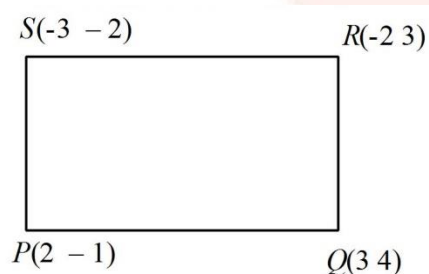
$$\begin{aligned} LHS &= \frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} \\ &= \frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \quad (\because \sin^2 \theta = 1 - \cos^2 \theta) \\ &= \frac{\sin \theta [1 - 2(1 - \cos^2 \theta)]}{\cos \theta (2 \cos^2 \theta - 1)} \\ &= \frac{\sin \theta [1 - 2 + 2 \cos^2 \theta]}{\cos \theta (2 \cos^2 \theta - 1)} \\ &= \frac{\sin \theta (2 \cos^2 \theta - 1)}{\cos \theta (2 \cos^2 \theta - 1)} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta \\ &= RHS \end{aligned}$$

Hence LHS = RHS.

28. If there are four points P(2, -1), Q(3, 4), R(-2, 3) and S(-3, -2) in a plane, then prove that PQRS is not a square but a rhombus.

Solution:

Consider



$$\begin{aligned}PQ &= \sqrt{(3-2)^2 + (4+1)^2} \\&= \sqrt{1+25} \\&= \sqrt{26} \text{-----(i)}\end{aligned}$$

$$\begin{aligned}QR &= \sqrt{(3+2)^2 + (4-3)^2} \\&= \sqrt{25+1} \\&= \sqrt{26} \text{-----(ii)}\end{aligned}$$

$$\begin{aligned}RS &= \sqrt{(-2+3)^2 + (3+2)^2} \\&= \sqrt{1+25} \\&= \sqrt{26} \text{-----(iii)}\end{aligned}$$

$$\begin{aligned}SP &= \sqrt{(2+3)^2 + (2-1)^2} \\&= \sqrt{25+1} \\&= \sqrt{26} \text{-----(iv)}\end{aligned}$$

Equation 1, 2, 3 and 4 are equal

$$PQ = QR = RS = SP = \sqrt{26}$$

Then,

$$\begin{aligned}PR &= \sqrt{(2+2)^2 + (3+1)^2} \\&= \sqrt{16+16} \\&= \sqrt{32} \text{-----(5)}\end{aligned}$$

$$\begin{aligned}QS &= \sqrt{(3+3)^2 + (4+2)^2} \\&= \sqrt{36+36} \\&= \sqrt{72} \text{-----(6)}\end{aligned}$$

Equation (5) and (6)

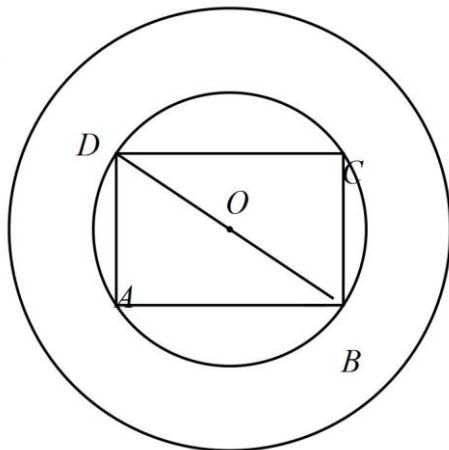
$$PR \neq QS$$

Hence, the PQRS is not a square but a rhombus.

29. Prove that the opposite angles of cyclic quadrilateral are supplementary or sum is 180° .

Solution:

Solution.



$$\angle BAD + \angle BCD = 180^\circ$$

$$x + y = 360^\circ \dots\dots(i)$$

$$\angle BAD = \frac{1}{2}y \dots\dots(ii)$$

$$\angle BCD = \frac{1}{2}x \dots\dots(iii)$$

adding equation(ii) and equation (iii)

$$\angle BAD + \angle BCD = \frac{1}{2}y + \frac{1}{2}x = \frac{y+x}{2} = 180^\circ$$

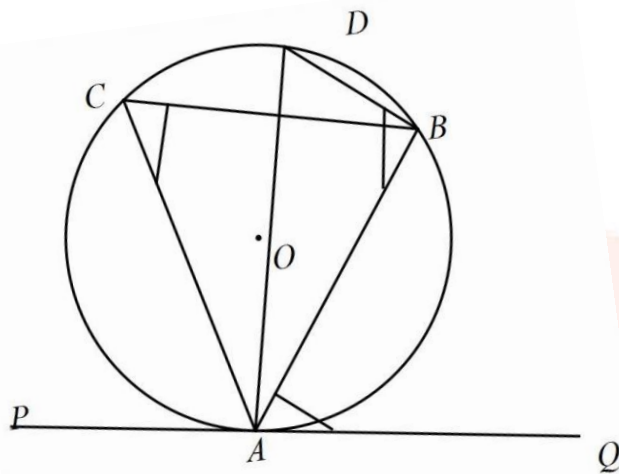
Hence, prove it.

OR,

Prove that if a chord is drawn from a point of contact of the tangent of the circle then angle made by this chord with the tangent are equal to the respective alternate angles made by segments with this chord.

Solution:

Construction – construction AD of diameter to point A.



$\angle ACB = \angle ADB$(i) circle of segment are equal angel

$\angle ABD = 90^\circ$(ii) make a angle of semi-circle

$\angle DAQ = 90^\circ$(iii) $\therefore DA \perp PQ$

$\triangle ABD$

$\angle ABD + \angle BAD + \angle ADB = 180^\circ$

$\angle BAD + \angle ADB = 90^\circ$

$\angle BAD + \angle ADB = \angle DAQ$

$\angle BAD + \angle ADB = \angle BAD + \angle BAQ$

$\angle ADB = \angle BAQ$

$\angle ACB = \angle QAB$ equation (i)

Hence, prove it.

30. Find the median and mode of the following frequency distribution.

Class	10-25	25-40	40-55	55-70	70-85	85-100
f_i	6	20	44	26	3	1

Solution:

Class	Frequency f_i	Cumulative frequency
10-25	6	6
25-40	20	26
40-55	44	70
55-70	26	96
70-85	3	99
85-100	1	100
	$\sum f_i = 100$	

Now,

$$N = 100 \Rightarrow \frac{N}{2} = \frac{100}{2} = 50$$

The cumulative frequency just greater than 44 is 70 and the corresponding class is 40-55

$$l = 40, h = 15, f_1 = 44, f_0 = 20$$

$$\text{Mode}(z) = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$= 40 + \left(\frac{44 - 20}{88 - 20 - 26} \right) \times 15$$

$$= 40 + \left(\frac{24}{88 - 46} \right) \times 15$$

$$= 40 + \frac{24}{42} \times 15$$

$$= 40 + \frac{60}{7}$$

$$= 40 + 8.57$$

$$= 48.57$$

$$l = 40, h = 15, cf = 26, f = 44$$

$$l = 40, h = 15, cf = 26$$

$$\text{Median, } (m) = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$= 40 + \left(\frac{50 - 26}{44} \right) \times 15$$

$$= 40 + \frac{24}{44} \times 15$$

$$= 40 + \frac{90}{11}$$

$$= 40 + 8.18$$

$$= 48.18$$